B.TECH COMPUTER SCIENCE & ENGINEERING • PROJECT EVALUATION

SyncTube: Real-Time Synchronized Watch Party Platform

Simple Questions • Short Answers • Practical Architecture & Live Workflow

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1. PROBLEM DEFINITION

What Problem Are We Solving?

Modern video platforms are built for 1 person alone • Shared watching is broken

THE PAIN POINTS TODAY

• **Isolated Streaming:** YouTube and OTT platforms isolate every viewer into their own unicast stream.

• **Manual 3-2-1 Play Fails:** Buffer variations, network jitter, and hardware clock drift cause desynchronization within minutes.

• **Mid-Roll Commercial Ads:** Commercials play at different times for different users, completely breaking synchronization.

• **Algorithmic Doomscrolling:** Clickbait sidebars and addictive Shorts distract students trying to watch study lectures.

HOW SYNCTUBE FIXES IT

• **Authoritative Synchronization:** Sub-100ms playout alignment. Play, pause, and seek happen simultaneously for everyone.

• **Zero Ad Interruptions:** Videos play cleanly without disruptive commercial ads stopping the lecture flow.

• **Pure Cinema Interface:** Strips away algorithmic feeds, recommendations, and shorts for total study focus.

• **Zero Barriers to Entry:** No account creation, no password fatigue, and no third-party browser extensions needed.

2. TARGET AUDIENCE

Who Is This Platform Built For?

Tailored specifically for three high-demand everyday user demographics

🎓 STUDENTS & STUDY GROUPS

• **Playlist Import:** Paste entire YouTube course playlists (Gate Smashers, NPTEL, MIT OCW).

• **Ad-Free Learning:** Zero mid-roll commercial interruptions.

• **No Distractions:** Stripped of Shorts and clickbait feeds.

• **Discussion Pauses:** Pause together to discuss difficult concepts and take notes.

❤️ LONG-DISTANCE COUPLES

• **Virtual Date Nights:** Watch romantic movies, series, and vlogs together across cities.

• **Emotional Sync:** Laugh and react at the exact same millisecond with sub-100ms sync.

• **Video Calling:** Face-to-face WebRTC video & audio while watching.

• **Floating Hearts:** Send animated live floating reaction emojis.

🍿 FRIENDS & WATCH PARTIES

• **Weekend Marathons:** Anime, gaming highlights, comedy sketches, and movie nights.

• **Democratic Queue:** Everyone adds videos and upvotes their favorites.

• **Live Chat Banter:** Threaded messaging with emoji reactions.

• **Soundboard:** Trigger synchronized funny sound effects.

3. SYSTEM WORKFLOW

How Does SyncTube Work?

A seamless 4-step user journey from zero to watching together in under 5 seconds

STEP 1

Create a Room in 2s

Enter your name, pick an optional anime avatar, and click Start Watch Party. No login needed.

STEP 2

Instant Share & Join

Share the auto-generated 6-character room code, direct invite link, or display the scannable QR code.

STEP 3

Load Video / Playlist

Paste any YouTube link, import an entire lecture playlist, or search videos directly in the app.

STEP 4

Enjoy Synchronized Media

Play, pause, seek, chat, vote on the queue, or start video calling. The server coordinates state instantly.

4. SYNCHRONIZATION MECHANISM

How Do We Keep Everyone in Exact Sync?

Authoritative clock offset calculation + 3-Tier Adaptive Drift Guard

CORE SYNCHRONIZATION PRINCIPLE

“SyncTube synchronizes the watch experience, not the video stream.”

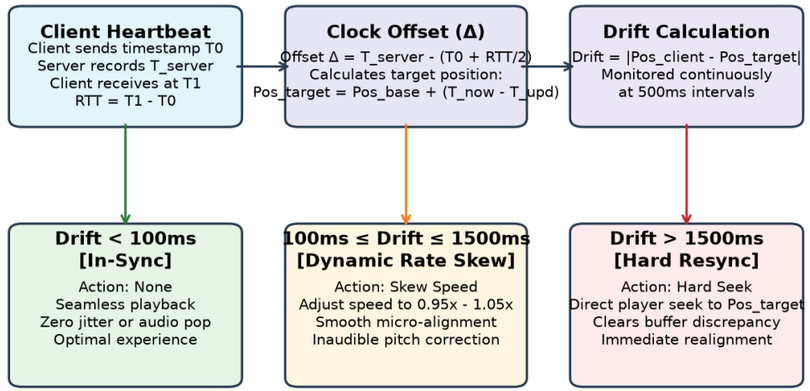
• **NTP Clock Offset Calculation:** Clients periodically exchange timestamped heartbeat probes (T0, T_server, T1) to calculate Round-Trip Time (RTT) and network latency.

• **Offset Formula:** Δ = T_server - (T0 + RTT / 2). Every client accurately knows the true server clock position.

• **Target Playhead:** Pos_target = Pos_base + (T_now + Δ - lastUpdated).

• **Zero Stream Proxying:** Video frames are rendered natively via YouTube IFrame API; server only sends tiny JSON timeline sync events.

Figure 4.1: NTP-Like Clock Synchronization & Adaptive Drift Guard State Machine



5. ACADEMIC USE-CASE

Why Is SyncTube a Game-Changer for Students?

Eliminating distractions and commercial interruptions from group study

FOCUSED GROUP STUDY & LECTURES

• **Bulk Playlist Import:** Students can paste entire course playlists (OS, DBMS, DSA). All lectures queue chronologically.

• **Zero Commercial Ads:** Commercials that interrupt professors mid-explanation are completely eliminated.

• **Distraction-Free Cinema UI:** Strips away recommended feeds, clickbait thumbnails, and addictive Shorts.

COLLABORATIVE LEARNING TOOLS

• **Synchronous Doubt Pausing:** When any student pauses to understand a diagram, it pauses for everyone.

• **In-Room Note Sharing:** Students share time-stamped notes, code snippets, and questions directly in chat.

• **Peer Accountability:** Studying together in a virtual room creates healthy study discipline similar to a library.

6. SOCIAL & RELATIONSHIP USE-CASES

Why Do Couples & Friends Love It?

Recreating the warmth of physical co-presence across distance

❤️ FOR LONG-DISTANCE COUPLES

• **Zero Spoilers:** Sub-100ms sync ensures partners react to emotional twists and jokes at the exact same physical moment.

• **Face-to-Face Video Calling:** Integrated WebRTC calling lets partners see each other live right next to the player.

• **Floating Emoji Reactions:** Heart and celebration particles float over the screen to express real-time affection.

🍿 FOR FRIENDS & WATCH PARTIES

• **Democratic Queue Voting:** Everyone queues songs/clips; the group upvotes, and top songs play next automatically.

• **Live Chat Replies & Banter:** Full threaded messaging for funny commentary and group jokes.

• **Soundboard Effects & Polls:** Trigger comedy sound effects and create instant group polls.

7. ARCHITECTURAL SANITY

What About Installation, Privacy & Security?

Zero installation, zero DRM circumvention, and zero data harvesting

ZERO BROWSER EXTENSIONS

Runs purely in standard web browsers (Chrome, Edge, Firefox, Safari). No extension installation that scares non-technical users or gets blocked by college IT firewalls.

ZERO MANDATORY SIGN-UP

Users spin up or join a room in seconds with only a display name and room code. Eliminates account creation fatigue and prevents email spam.

ZERO DRM VIOLATIONS

SyncTube adheres strictly to copyright law. We do not download, proxy, or re-encode video streams. Video rendering is delegated directly to the official YouTube IFrame player.

STRICT BACKEND VALIDATION

All WebSocket payloads and REST endpoints pass through runtime Zod schema validation, blocking malformed event injections and protecting room states.

8. COMPETITIVE MATRIX

Why Not Just Use Existing Services?

Direct comparison across key usability, technical, and feature dimensions

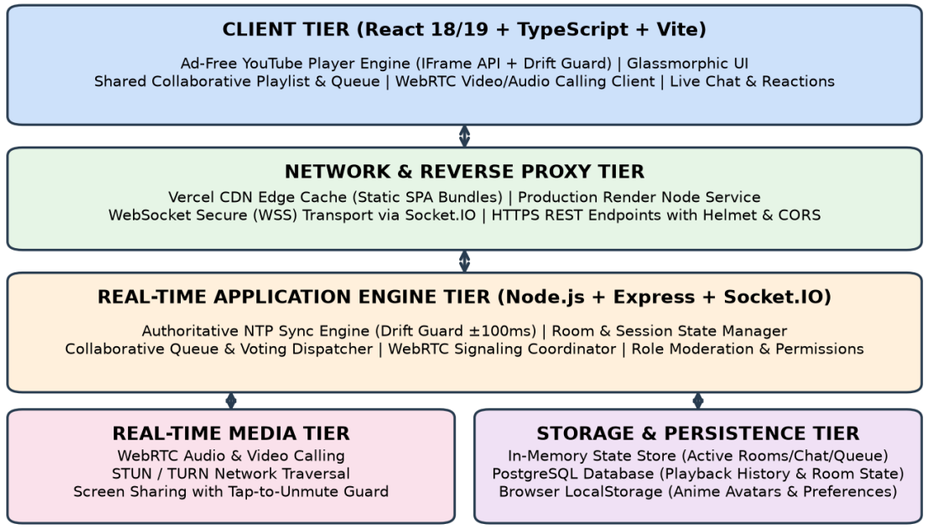
Feature / Dimension	SyncTube (Our System)	Teleparty (Netflix Party)	Watch2Gether	Discord Watch Together
Installation Required	**Zero Install (Web Native)**	Mandatory Extension	Zero Install (Web Native)	Discord App Required
Account Creation	**Optional (Instant Code)**	Mandatory Sign-Up	Optional	Mandatory Discord Acct
Ad-Free Cinema Stage	**Yes (No Ads / No Feeds)**	No (Standard UI)	No (Heavy Banner Ads)	No (Discord Overlay)
Playlist Queue & Voting	**Full Import & Upvoting**	No Playlist Import	Basic Manual Queue	Basic Queue Only
Playout Sync Precision	**Sub-100ms (NTP + Skew)**	300ms – 1200ms	250ms – 800ms	150ms – 500ms
Built-in Video Calling	**WebRTC Video/Audio Call**	Text Chat Only	Laggy P2P Mesh	Discord Voice/Video
Mobile Phone Usability	**Dedicated Touch Dock**	No Mobile Support	Limited Mobile Web	Mobile App Only

9. SYSTEM ARCHITECTURE

System Architecture & Engineering Stack

Built with modern, production-tested full-stack web technologies

Figure 3.1: SyncTube High-Level Multi-Tier System Architecture



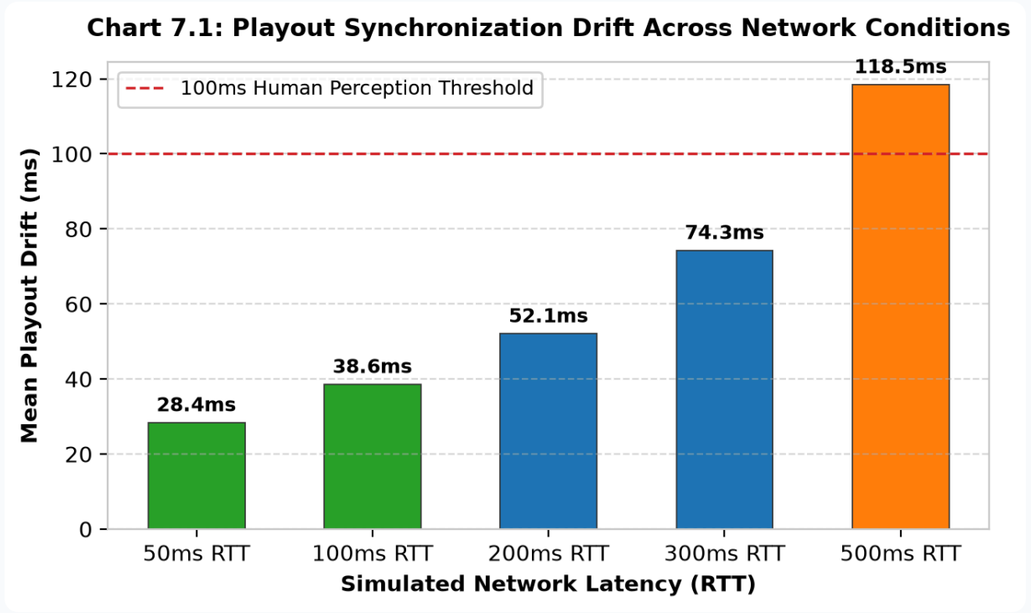
CORE TECHNOLOGY STACK

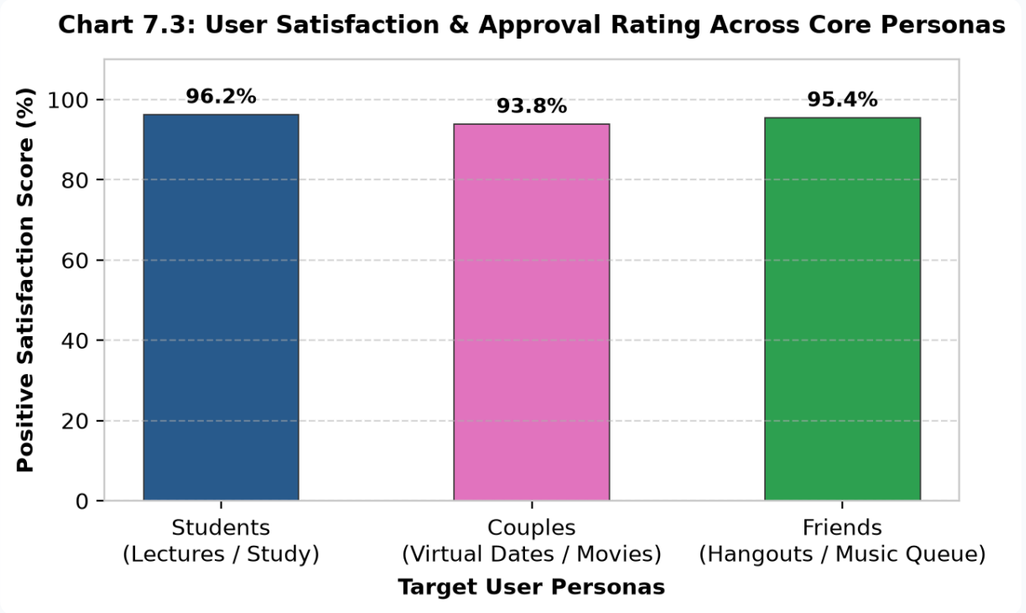
- **Client Frontend:** React 18/19, TypeScript, Vite, Glassmorphic CSS with ambient player lighting.
- **Real-Time Engine:** Node.js, Express, Socket.IO v4 full-duplex WebSocket bus with Zod schema validation.
- **Media Calling:** WebRTC (RFC 8825) with Google STUN and configurable TURN traversal.
- **Data Tier:** In-Memory volatile store (sub-millisecond room state) + PostgreSQL history database.
- **Cloud Edge:** Deployed on Vercel CDN (Frontend) and Render Cloud (Backend Server).

10. RESULTS & BENCHMARKS

What Makes the Project Successful?

Empirical performance testing and user satisfaction metrics





11. SUMMARY & ROADMAP

Our Core Pitch & Future Roadmap

Transforming isolated video watching into a shared, productive, and fun social reality

🎯 OUR CORE PITCH

“SyncTube brings people together to watch YouTube in perfect real-time sync. With zero ads, zero algorithmic distractions, instant 6-character room codes, collaborative playlist voting, and integrated video calling, it is the ultimate shared viewing platform for students, couples, and friends.”

🚀 FUTURE SCOPE & ENHANCEMENTS

- **Multi-Region Redis Pub/Sub Cluster:** Scale room state horizontally across global cloud servers for international watch parties.
- **Progressive Web App (PWA):** Enable mobile background audio and Picture-in-Picture (PiP) support for smartphones.
- **AI Real-Time Subtitles:** On-device speech-to-text generating real-time multilingual subtitles for educational lectures.